RUIWEN ZHOU

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EDUCATION

Ph.D. student in Computer Science, National University of Singapore — Aug 2025 – Present

M.Eng. in Computer Science, Shanghai Jiao Tong University — Sep 2022 – Mar 2025

B.Eng. in Information Engineering, Shanghai Jiao Tong University — Sep 2018 – Jun 2022

RESEARCH INTERESTS

My research generally focuses on LLM Agents, especially how to acquire the necessary context and utilize it effectively for long-horizon tasks. Specifically, I have been working in the following areas recently:

- LLM mid-training and post-training.
- Context management and memory augmentation.
- LLM-based multi-agent systems.

RESEARCH EXPERIENCE

Ph.D. Student — Aug 2025 – Present
WING Lab, National University of Singapore
Advisor: *Prof. Min-Yen Kan & Prof. Soujanya Poria*

Visiting Student — Jul 2024 – Dec 2024
NLP Group, UC Santa Barbara
Advisor: *Prof. William Yang Wang*

Student Researcher — Jan 2021 – Mar 2025
APEX Lab, Shanghai Jiao Tong University
Advisor: *Prof. Weinan Zhang*

INTERNSHIP EXPERIENCE

Research Intern (Top Talent Program) — Apr 2026 – Present
MiniMax AI

- **Agent Spec Mid-training for Value Alignment:** MiniMax M2.7 models show poor alignment with values including honesty, resilience, etc. Based on the model spec mid-training method, we synthesize documents that involve these properties and connect them to models' identity and use these documents in the CPT process. Experiments show that this process significantly improves model alignment in our internal benchmark, and this process has been merged into the continual pre-training (CPT) pipeline of MiniMax M3 / M3.1.
- **Agent Belief Diagnosis in Long-Horizon Tasks:** To gain more insights into the root causes of agents' failures, we are building an automatic diagnosis pipeline to locate where agents make mistakes about its completion status of task requirements and awareness about relevant facts. We have found through probing that mainstream LLMs are likely to have incorrect beliefs about such information in certain cases and we will further examine whether LLMs can act correctly given correct beliefs. The findings will help the training of future models.

Research Intern — Feb 2022 – Feb 2023
Amazon Web Services (AWS)

- **Tabular Data Prediction with Retrieval Augmentation:** Previous tabular data prediction methods focused on column-wise feature interaction within a single row, without utilizing information from similar samples. Therefore, we propose a retrieval-augmented method PET, which retrieves relevant samples with the most influential features and utilizes an attention-based model to aggregate the information. Experiments show that PET significantly surpasses single-row baselines with very little computation overhead.

Research Intern Aug 2021 – Jan 2022
Microsoft Research Asia (MSRA)

- **Risk-Sensitive RL with Trajectory Q-Learning:** Previous risk-sensitive RL methods cannot solve the policy optimization problem correctly with most distortion risk measures as targets. Through theoretical analysis we reveal that the issue comes from the Markovian Bellman update, which drops the distribution over history returns. To address this issue, we model the value function as the distribution of accumulated return over the whole trajectory rather than only the future steps, and modify the Bellman update formula accordingly. The proposed Trajectory Q-Learning method can theoretically converge to the optimal value and practically outperform Markovian baselines.

RESEARCH PROJECTS

Epistemic Context Learning: Learning to Trust in Multi-Agent Systems *Under Review*
WING Lab, NUS

- Investigated LLMs' sycophancy and inadequate ability to evaluate peer reliability in multi-agent systems, and formulated the learning problem of *history-aware reference* as an alternative solution, which introduces the historical interactions of peers as additional input and shifts the task from evaluating peer reasoning quality to estimating peer reliability based on interaction history.
- Developed *Epistemic Context Learning* (ECL), a two-stage framework that estimates peer reliability from interaction history and conditions answer aggregation on explicit peer profiles. We further proposed to optimize models by GRPO training using auxiliary rewards based peer identification accuracy.
- Conducted extensive experiments to show that ECL significantly improves the reasoning of both small open-source LLMs and large close-source LLMs, and did additional analysis that provides insights on how historical trust is learned and utilized.

Measuring and Mitigating Rapport Bias in Multi-Agent Interactions *ICLR 2026*
WING Lab, NUS

- Investigated the vulnerability of LLMs to social rapport bias and peer pressure, which often compromises factual accuracy in group settings.
- Contributed to construction of KAIROS benchmark and evaluated GRPO with outcome rewards as a mitigation strategy, validating that RL-tuned LLMs can improve over base models in resisting social conformity.

RuleArena: Rule-Guided Reasoning with LLMs in Real-World Scenarios *ACL 2025 (Main)*
NLP Group, UCSB

- Addressed the limitation of existing benchmarks that focus on simple first-order logic, failing to evaluate LLMs' adherence to complex, real-world constraints (e.g., tax/legal rules).
- Led the creation of RuleArena, a benchmark requiring long-context understanding and multi-hop reasoning based on real-world applications (airlines, NBA, and tax). Developed rigorous evaluation metrics to decouple rule retrieval errors from rule reasoning errors.
- Conducted comprehensive evaluation of mainstream LLMs (GPT-4o, Claude-3.5, etc.), revealing critical gaps in handling complex real-world logic constraints compared to first-order logic tasks.

TRAD: Step-Wise Thought Retrieval and Aligned Decision for LLM Agents *SIGIR 2024 (Oral)*
APEX Lab, SJTU

- Addressed the issue of "plausible but wrong" and noisy in-context example in trajectory-level retrieval for long-horizon agentic tasks.
- Developed TRAD, a novel agent framework combining Step-wise Thought Retrieval (using CoT as search queries and keys) and Aligned Decision (for concentrating on key trajectory segments) mechanisms.
- Achieved SOTA performances on ALFWorld and Mind2Web. Successfully deployed in enterprise RPA scenarios at China Pacific Insurance Company (CPIC), improving task success rate from 65.0% to 92.5%.

SELECTED PUBLICATIONS

[1] **Ruiwen Zhou**, Maojia Song, Xiaobao Wu, Sitao Cheng, Xunjian Yin, Yuxi Xie, Zhuoqun Hao, Wenyue Hua, Liangming Pan, Soujanya Poria, Min-Yen Kan. Epistemic Context Learning: Building Trust the Right Way in LLM-Based Multi-Agent Systems. *arXiv preprint arXiv:2601.21742*.

[2] **Ruiwen Zhou**, Wenyue Hua, Liangming Pan, Sitao Cheng, Xiaobao Wu, En Yu, William Yang Wang. RuleArena: A Benchmark for Rule-Guided Reasoning with LLMs in Real-World Scenarios. *ACL 2025 (Main)*.

[3] **Ruiwen Zhou**, Yingxuan Yang, Muning Wen, Ying Wen, Wenhao Wang, Chunling Xi, Guoqiang Xu, Yong Yu, Weinan Zhang. TRAD: Enhancing LLM Agents with Step-Wise Thought Retrieval and Aligned Decision. *SIGIR 2024 (Oral)*.

[4] Shengtao Zhang, Jiaqian Wang, **Ruiwen Zhou**, Junwei Liao, Yuchen Feng, Weinan Zhang, Ying Wen, Zhiyu Li, Feiyu Xiong, Yutao Qi, Bo Tang, Muning Wen. MemRL: Self-Evolving Agents via Runtime Reinforcement Learning on Episodic Memory. *NeurIPS 2026*.

[5] Junwei Liao, Haoting Shi, **Ruiwen Zhou**, Jiaqian Wang, Shengtao Zhang, Wei Zhang, Weinan Zhang, Ying Wen, Zhiyu Li, Feiyu Xiong, Bo Tang, Muning Wen. MemQ: Integrating Q-Learning into Self-Evolving Memory Agents over Provenance DAGs. *arXiv preprint arXiv:2605.08374*.

[6] Sitao Cheng, Xunjian Yin, **Ruiwen Zhou**, Yuxuan Li, Xinyi Wang, Liangming Pan, William Yang Wang, Victor Zhong. From Atomic to Composite: Reinforcement Learning Enables Generalization in Complementary Reasoning. *arXiv preprint arXiv:2512.01970*.

[7] Maojia Song, Tej D. Pala, **Ruiwen Zhou**, Weisheng Jin, Amir Zadeh, Chuan Li, Dorien Herremans, Soujanya Poria. Measuring and Mitigating Rapport Bias of LLMs under Multi-Agent Social Interactions. *ICLR 2026*.

[8] Xiaobao Wu, Liangming Pan, Yuxi Xie, **Ruiwen Zhou**, Shuai Zhao, Yubo Ma, Mingzhe Du, Rui Mao, Anh Tuan Luu, William Yang Wang. AntiLeak-Bench: Preventing Data Contamination by Automatically Constructing Benchmarks with Updated Real-World Knowledge. *ACL 2025 (Oral & SAC Highlight)*.

SELECTED AWARDS

- NUS Graduate Research Scholarship 2025
- Huatai Securities Fellowship 2024
- First-Class Excellence Scholarship 2022, 2024
- Zhiyuan Honors Scholarship (Top 5%) 2019, 2020, 2021
- China National Scholarship (Rank 1/144) 2020
- A-Class Excellence Scholarship (Rank 1/144) 2020